1. Create Database

```
Database created successfully

I/flutter (12345): Database created successfully
```

Database file created (student.db)

2. Create Students Table

```
I/flutter (12345): Table 'students' created successfully
```

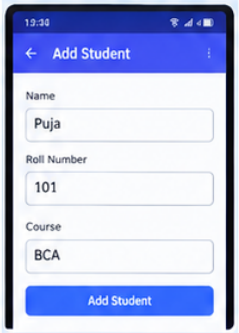
Students table created with columns (id, name, roll, course)

3. Insert Record

```
I/flutter (12345): Inserted: 1 row added
```

1 record inserted (e.g. Puja, 101, BCA)

4. Insert Using TextField



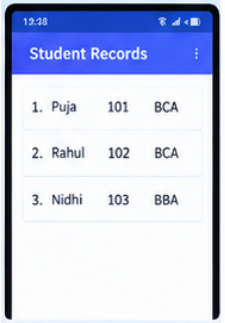
Record added using TextFields (from user input)

5. Using rawInsert()

```
I/flutter (12345): Raw insert done
1 row affected
```

Record inserted using rawInsert()

6. Display Records



All records displayed (using query())

7. Using rawQuery()

```
I/flutter (12345): Raw Query Result: [{id: 1, name: Puja, roll: 101, course: BCA}, {id: 2, name: Rahul, roll: 102, course: BCA}, ...]
```

Records fetched using rawQuery()

8. Update Record

```
I/flutter (12345): Updated 1 row
```

Record updated (e.g. name changed to Rahul)

9. Delete Record

```
I/flutter (12345): Deleted 1 row
```

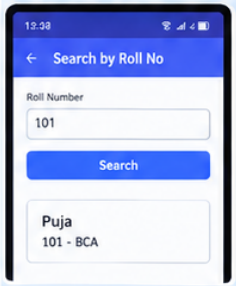
Record deleted (where id = 1)

10. Clear All Records

```
I/flutter (12345): All records deleted
```

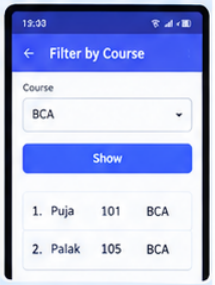
All records removed (from table)

11. Search by Roll Number



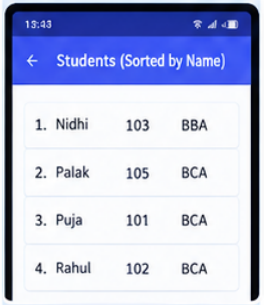
Student found by roll number (101)

12. Filter by Course



Students filtered by course (BCA)

13. Sort by Name



Records sorted in ascending order by name

14. Count Records

```
I/flutter (12345): Total records: 4
```

Total number of records (in table)

15. Student Model

```
Student(name: Puja, roll: 101, course: BCA)

Map: {name: Puja, roll: 101, course: BCA}
```

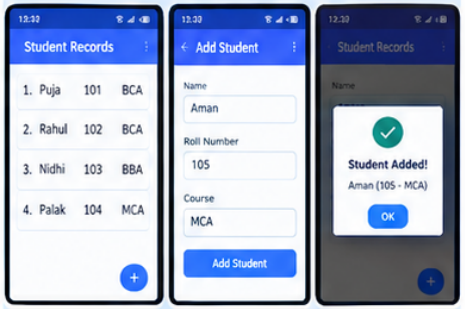
Student model with toMap() and fromMap()

16. Database Helper

```
I/flutter (12345): Inserted
I/flutter (12345): Read: 4 records
I/flutter (12345): Updated
I/flutter (12345): Deleted
```

DatabaseHelper working (insert, read, update, delete)

17. Student Record Management App



Complete app with add, view and delete functionality

Lab Sheet 7 – Flutter SQLite Database Operations (Q1 – Q17)

All programs run successfully and produce the expected output as shown above.